

Zeth Green

SENIOR PYTHON ENGINEER

Thanks for reading my CV. I love new projects, rescuing old legacy systems, meeting new people and learning about different industries. I'm usually hired as a Senior Python Engineer, but I'm happy to pitch in with whatever tool the task requires. Please get in touch! I don't bite.

Skills

Programming	Python, JavaScript, C/C++, and any other languages as needed.
Web	Django, FastAPI, Flask, asyncio, Twisted, React, HTML5, Pydantic, TypeScript, Lit.
AI	LangChain, Pydantic AI, Vertex AI, Google AI Platform, Kubeflow, LLMs.
Cloud/Engineering	AWS, GCP, Docker, Kubernetes, GitHub Actions, Buildbot, CI/CD, Linux, Windows, macOS.
Databases/Dataframes	PostgreSQL, MongoDB, MySQL, Berkeley DB, SQLite, DynamoDB, Pandas, NumPy.
Writing	Software Documentation, Copy Editing, Proofreading, Typesetting, Technical Review, Grant Writing.
Leadership	Coach, Tutor, Team Motivator, Scrum Master, Testing, Teamwork, Independent Working, Smiling.
Other	Full clean UK manual driving licence.

Experience

Turner & Townsend (via eSynergy / Zeth LTD)

Remote / occasional St Paul's

SENIOR SOFTWARE ENGINEER (CONTRACTOR)

October 2025 – Present

- Developed internal and client-facing software for cost control, capital planning & portfolio reporting for infrastructure and construction projects.
- Designed an append-only event ledger enabling accurate historical financial reporting as project portfolios changed.
- Delivered draft cost workspaces in Flask and Angular, allowing cost managers to prepare and approve changes before publishing.
- Built cashflow reporting comparing baseline and forecast expenditure, with robust handling of financial edge cases.
- Cut FastAPI client-portal search and filtering times from minutes to milliseconds by optimising database queries and indexing.
- Resolved asynchronous SQLAlchemy loading failures and task lifecycle issues in FastAPI services, improving reliability.
- Technologies: Python, Flask, FastAPI, SQLAlchemy, PostgreSQL, TypeScript, Angular, Ionic, Google Cloud, Docker and Kubernetes.

Zeth LTD

Remote

SENIOR SOFTWARE ENGINEER/DIRECTOR

February 2025 – September 2025

- Web development, writing, AI learning/research (text-corpus RAG projects, Pydantic-AI/LangChain), and conference participation.

Kingfisher PLC (Owner of B&Q, Screwfix, Castorama, Brico Dépôt) - Group Data Science

Remote / occasional Paddington

SENIOR MACHINE LEARNING PLATFORM ENGINEER (DJANGO CONTRACTOR)

March 2024 – January 2025

- Built a Django/React pricing interface for AI and machine learning models. Worked with end users and stakeholders.
- Used Python, Django, React, Kubernetes, Kubeflow, PostgreSQL, Grafana, GCP, BigQuery and Vertex AI.

ManyPets

Remote / occasional Farringdon

SENIOR PYTHON DEVELOPER (CONTRACTOR)

2021 – 2023

- Led a project to rewrite serverless payments software (collecting premiums and paying claims) to support expansion into the US and EU.
- Used Python, AWS Lambda, CloudWatch, Flask, and Stripe API. Django user web portal; OpenAPI documentation using Swagger.

Radius Payment Solutions

Crewe / occasional Manchester

SENIOR FULL STACK DEVELOPER (CONTRACTOR)

March 2018 – March 2021

- Telematics and fleet management applications using Django REST and React. Front-end testing using Jest and Enzyme.
- Led monthly all-hands developer meetings. Coached junior developers. Migrated the application to a full-stack Docker and AWS deployment pipeline.

Zeth LTD

Birmingham

FREELANCE

February 2017 – August 2017

- Full-stack web development across Python, JavaScript, back-end services, system administration, testing and training.

CGscope

Old Street, London

DJANGO DEVELOPER

March 2017 – July 2017

- Data mining and management tool for corporate governance intelligence analysts in the banking sector.

University of Birmingham

Edgbaston, Birmingham

LEAD SOFTWARE DEVELOPER, ESTORIA DE ESPANNA DIGITAL

July 2014 – February 2017

- Created Python software for researchers to record, analyse, compare and publish historical Spanish manuscript data.
- The web app featured on Spanish TV news and was widely applauded by scholars of medieval Spain.

Infomachine

Small Heath, Birmingham

SOFTWARE DEVELOPER

July 2015 – November 2016

- Worked on apps for Land Rover, a museum and a car parts distributor.

University of Birmingham

RESEARCH FELLOW, COLLEGE OF ARTS AND LAW

Edgbaston, Birmingham

June 2008 – July 2014

- Created web applications and Python libraries for University research data.
- Designed an asynchronous text-collation system handling tens of millions of manuscript variants. Maintained a Linux cluster.
- Advised on technical appendices to grant applications. Taught a Digital Humanities module covering Python programming and heritage app development.
- Collaborated in EU projects “Interedition” and “CollateX” to create editing software: <https://collatex.net/about/>

Scholarly Digital Editions Limited

WEB & XML DEVELOPER

Leicester

June 2008 – July 2009

- Software development for the critical digital edition of the medieval manuscripts of Dante’s Commedia.
- Processed XML data in C and TCL; formatted the HTML output in JavaScript. Website and best-selling academic CD-ROM.

Clockwork Software Systems

WEB APPLICATIONS DEVELOPER

Erdington, Birmingham

June 2007 – June 2008

- Created the first Personalised Registration Auction system for the DVLA. Real-time Django web app, synced with live in-person auction.

University of Birmingham

WEB EDITOR, BIRMINGHAM BUSINESS SCHOOL

Edgbaston, Birmingham

July 2006 – June 2008

- Built web services, online advertising, and a student-feedback system measuring performance of lecturers.

University of Birmingham

HELPPDESK WEBMASTER

Edgbaston, Birmingham

July 2005 – June 2006

- Wrote and compiled a website of online technical documentation for users of University computing services.

University of Birmingham

COMPUTING HELP ADVISOR

Edgbaston, Birmingham

October 2004 – July 2005

- Provided front-line support to students and staff for any computing problems or queries that they encountered.

Various (Freelancing)

WEB DESIGNER

Selly Oak, Birmingham

October 2002 – October 2004

- Fully funded my second degree by building web apps for small businesses on a short-term contract/freelance basis.

University of Essex

STUDENT SUPPORT

Wivenhoe

July 2001 – September 2002

- Provided a confidential listening and advice service to students. Saved the University tens of thousands of pounds by reducing drop-out rates.

Great Eastern Trains

TRAIN CREW

Ipswich to London Liverpool Street

October 1999 – June 2001

- Completely funded first degree with a Saturday and holiday job selling sandwiches and putting out disabled ramps.

Christian Salvesen

CONVEYOR OPERATOR

Oulton Broad, Suffolk

June 1999 – September 1999

- Set up and moved factory machinery for processing different vegetables. Operated near-infrared optical sorter.

Education

University of Birmingham

MASTER OF ARTS (MA), NATURAL LANGUAGE PROCESSING

Birmingham

September 2006 – July 2008

University of Birmingham

BACHELOR OF ARTS (BA), DIGITAL HUMANITIES

Birmingham

September 2002 – July 2005

University of Essex

BACHELOR OF SCIENCE (BSc), ECONOMETRICS

Wivenhoe

September 1998 – July 2001

Professional Activities

- Building an open-source, AI-assisted personal knowledge platform - with cross-platform CI/CD and release engineering via GitHub Actions.
- Maintaining a cross-platform library for detecting hardware capabilities and classifying devices into practical performance tiers.
- Current research interest: computational epistemology, particularly in the AI context.
- Regular conference participation. 2025: DjangoCon Europe, Dublin (Apr; closed a Django bug in the sprint) and PyCon UK, Manchester (Sept).
- Exploring practical and creative applications of AI. Write stories. Play chess really badly.
- Founded Python West Midlands in 2006, which provides tutorials, talks and networking activities.
- Co-founded PyConUK, the UK’s Python conference, in 2007; vice-chair for seven years and chair for two years. EuroPython organiser, 2008–2010.
- Contributed to open source projects including Python. Created the Inputs module for cross-platform Python support for USB devices.
- Nominated and elected as a Fellow of the Python Software Foundation in 2010 for contributions to the Python community.